1. Cooling system and how a pressure sensor is used

Basic Physics of Cooling Battery and Pressure:
Bernoulli's Principle: Total energy = Static Pressure + density*velocity^2
(If the kinetic energy of the air increases, the potential energy of the static pressure must drop)

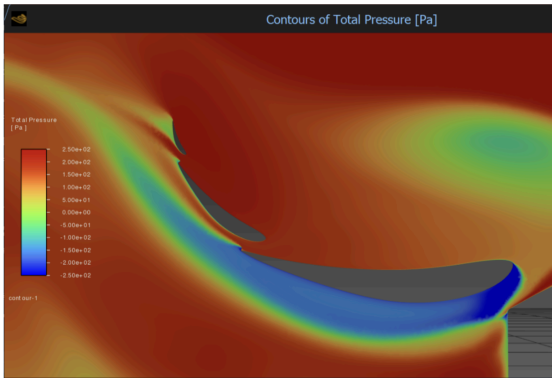
PV = nRT

Assuming an air-cooled accumulation system: an accumulator has a sensor to measure the volumetric pressure drop

- We need one sensor at the inlet, and another sensor at the outlet → **differential pressure sensor**

2. Pressure sensor Specifications that are important for this application:

1. **Range**: the smallest to largest value the sensor is capable of measuring.
2. **Resolution**: the smallest change is that the sensor can distinguis/report
Ex. 10005 Pa vs 10004 Pa if the sensor has a resolution of 1 Pa
3. **Accuracy**
4. **Precision**: if I measure repeatedly, how often do I get the same result?



Based on this graph, the entire aerodynamic phenomenon I am trying to measure is just 500 Pa wide.

- Atmospheric pressure is 101,325 Pa

3. Different Types of Sensors

Absolute vs. differential pressure sensors:
Absolute:

Differential pressure sensor:
could be used for things like intake restrictor monitoring (you have Port A and Port B; do we bring these 2 ports together?)

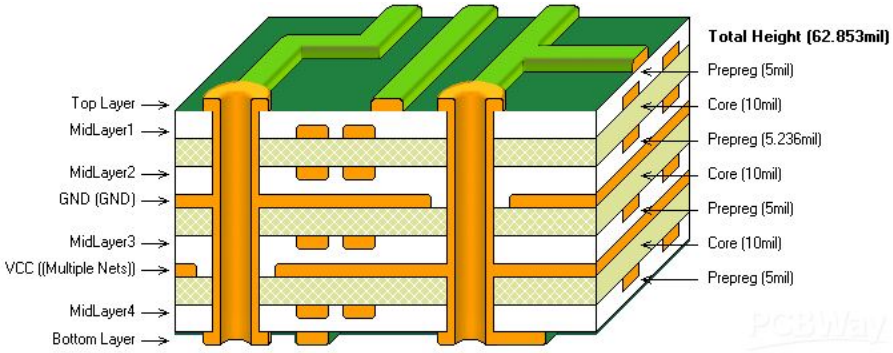
- **MEMS**: its mechanical pressure diaphragm is down to the micrometer level. High pressure pushes on the silicon diaphragm, and lower air pressure pushes up against the bottom. P1 > P2 → the diaphragm bends down a few nanometers → ASIC passes current through resistors and measures the change in voltage bc of the bending → converts it to Pa through a built-in calibration equation.
    - Extremely lightweight, take up a few square milimeters, measures pressure changes down to 0.1 Pa, chassis vibrations don't fake out this sensor bc the moving mass of its diaphragm is so low.
    - It has an onboard digital ASIC which allows you to chain multiple sensors along the I2C

4. Communication Interfaces for Sensors

Types of Communication interfaces and trade-offs

Type of Interface	Data rate	Distance	Power / Pins	Other
SPI				
I2C				
Onewire				

Analog				
UART				


Total Height (62.853mil)
Prepreg (5mil)
Core (10mil)
Prepreg (5.236mil)
Core (10mil)
Prepreg (5mil)
Core (10mil)
Prepreg (5mil)
Top Layer
MidLayer1
MidLayer2
GND (GND)
MidLayer3
VCC ([Multiple Nets])
MidLayer4
Bottom Layer